

## Lab 8

### Introduction to Rules and Queries in Prolog

Example problem

Facts:

- 1) Cow eats grass.
- 2) Sheep eats grass
- 3) Cat eats fish.
- 4) Dog eats meat.

Rules:

- 1) If an animal eats grass then the animal is herbivore.
- 2) If an animal eats meat then the animal is carnivore.
- 3) If an animal eats fish then the animal is carnivore.

Methodology:

- 1) Download the SWI-Prolog from the provided reference
- 2) Create a file with extension **.pl** and write down the above facts and rules in that file in the following format

*% These are the facts. They are the basic truths of our program.*

*eats(cow, grass).*

*eats(sheep, grass).*

*eats(cat, fish).*

*eats(dog, kibble).*

*eats(dog, meat).*

*% These are the rules. They allow Prolog to infer new information.*

*% A herbivore is an animal that eats grass.*

*is\_herbivore(Animal) :- eats(Animal, grass).*

*% A carnivore is an animal that eats meat or fish.*

*is\_carnivore(Animal) :- eats(Animal, meat).*

*is\_carnivore(Animal) :- eats(Animal, fish).*

- 3) Save the file.
- 4) Open SWI-Prolog

- 5) In the **file** menu, select **consult**, open the **.pl** file you just saved above. If the facts and rules are loaded correctly then you will see the rules file loaded message

```
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/users/avskh/Documents/Prolog/Rulebased/rules.pl ccopiled 0.00 sec, 8 clauses
?-
|
```

6)

Now you can insert your queries

- a) Let us first check the query *eats(cat, fish)*. Since the fact is present in our file, it should return true

- b) Similarly check for the following

i) *is\_herbivore(cow)*.

ii) *is\_carnivore(Animal)*.

(in case of multiple response we can see it as the following, **inserting a semicolon after the first response**)

### Report format

**Title:** Title as above

**Date:** Lab assigned date

### **Objective**

- To implement a rule based system using Prolog

### **Theory**

- Prolog programming language
- Rule based knowledgebase

### **Implementation**

- Write down the implementation steps as defined above
- Paste the screenshots of your outputs for provided queries

### **Conclusion**